

TECHNICAL ASSIGNMENT

RAG-Based Chatbot for Pernod Ricard

ISSUED BY	ROLE	DURATION
MarkGrid / Two99	AI Developer	72 Hours

TOTAL MARKS	INTERFACE	CORE RAG	GUARDRAILS
100 Points	10 Points	45 Points	20 Points

1. Overview

You are required to design and deliver a production-ready Retrieval-Augmented Generation (RAG) chatbot for Pernod Ricard, one of the world's largest premium spirits companies. The chatbot must function as an intelligent brand concierge, answering questions about Pernod Ricard's product portfolio, company history, sustainability initiatives, and responsible drinking guidelines — all grounded exclusively in publicly available web content.

This is not a prototype exercise. You are expected to demonstrate sound architectural decisions, clean and maintainable code, and all required guardrails implemented and verifiable. Evaluate your submission as if you are handing it to an enterprise client.

2. Assignment Tasks

Task 1 — Data Ingestion & Knowledge Base Construction

Scrape and ingest content from publicly available sources. Your knowledge base must cover the following areas:

- Brand portfolio: Absolut, Chivas Regal, Jameson, The Glenlivet, Beefeater, Ballantine's, Royal Salute, Malibu, Kahluá, Mumm, Perrier-Jouët, and all other major brands.
- Product details: Tasting notes, ABV, variants and expressions, geographic origin, production method, notable awards and accolades.
- Company information: History (founded 1975 via merger), headquarters (Paris), global presence, leadership, financial overview.
- Sustainability and CSR: 'Good Times from a Good Place' strategy, environmental targets, WISE responsible drinking programme.
- Trade and retail: Where to buy, duty-free presence, regional market context including India.

Acceptable sources: pernod-ricard.com, brand sub-sites (absolut.com, chivas.com, etc.), Wikipedia, Difford's Guide, Decanter, Reuters, FT. No hallucination — every response must trace to ingested content.

Task 2 — RAG Pipeline Architecture

Design and implement a clean, modular RAG pipeline. Minimum required components:

- Chunking strategy: Semantic or hybrid chunking — not naive fixed-size splitting. Justify your choice in the README.

- Embeddings: Use a production-grade model (e.g. OpenAI text-embedding-3-small, Cohere embed-v3, or a capable sentence-transformer). State your model and rationale.
- Vector store: Pinecone, Weaviate, Qdrant, ChromaDB, or FAISS. Justify your selection.
- Retrieval: Hybrid retrieval (dense + sparse / BM25) preferred. At minimum, dense retrieval with MMR re-ranking to prevent repetitive context.
- LLM layer: GPT-4o, Claude Sonnet 3.5, or Gemini 1.5 Pro. Do not use a model weaker than GPT-3.5-turbo-16k.
- System prompt: Must enforce role, brand tone, knowledge boundary, and all guardrails (see Task 3) at the prompt level.

Task 3 — Guardrails — Mandatory Implementation

Pernod Ricard is a regulated alcohol brand operating across multiple jurisdictions. All six MUST guardrails below are non-negotiable and must be demonstrable in your test suite.

GUARDRAIL	REQUIRED BEHAVIOUR	PRIORITY
Age Gate	The bot must confirm the user is of legal drinking age before any product discussion. Under-age users receive a hard redirect to responsible drinking resources.	MUST
No Sales / Pricing	The bot must never quote prices, enable purchases, or link to e-commerce checkout flows. Redirect to 'find a store' or the official brand website only.	MUST
Responsible Drinking	All cocktail recipe or consumption queries must include a responsible drinking reminder. Queries about dangerous consumption or intoxication must be firmly redirected.	MUST
Competitor Restriction	The bot must not compare Pernod Ricard products negatively against competitors (Diageo, Bacardi, Brown-Forman, etc.) or recommend competitor brands. Decline gracefully.	MUST
Medical / Legal Queries	Questions about alcohol and medication, addiction, or drink-driving must immediately redirect to qualified resources (e.g. NIAAA, Drinkaware, local helplines).	MUST
Hallucination Boundary	If retrieval yields no relevant context, the bot must state 'I don't have that information.' Fabricating product details, awards, or claims is strictly prohibited.	MUST
Off-topic Rejection	Political, unrelated commercial, personal, or harmful queries must be declined in one sentence and redirected to brand-relevant topics.	SHOULD
Brand Tone	All responses must reflect Pernod Ricard's premium voice — knowledgeable, warm, never salesy. No slang, no emojis, no casual disregard for brand identity.	SHOULD

Task 4 — User Interface

Build a functional chat interface suitable for client demonstration. This does not need to be pixel-perfect, but must be fully usable.

- Chat window with message history, input field, and send button.
- Source citations displayed alongside each bot response (document title or URL).
- Age gate confirmation on first session load.
- Loading or streaming state indicator for LLM response generation.
- Acceptable frameworks: Streamlit, Gradio, Next.js, or plain HTML/JS.

Bonus: Multi-turn session memory. Visual flagging of low-confidence responses (e.g. 'Based on limited available data').

Task 5 — Evaluation & Test Suite

Include a documented test suite with a minimum of 15 test queries, covering the following categories:

- Product knowledge: e.g. 'What are the expressions of The Glenlivet?'
- Brand history: e.g. 'When was Chivas Regal founded and what makes it distinctive?'
- Cocktail / serve: e.g. 'How do I make a classic Absolut Cosmopolitan?'
- Guardrail triggers: Underage simulation, competitor comparison attempt, medical query, pricing query — each must produce a compliant refusal.
- Edge cases: Out-of-scope topic, ambiguous query, empty retrieval result.

For each test, log: query → retrieved chunks → final response → pass / fail outcome against expected guardrail or content result.

Optional but valued: RAGAS-based automated evaluation reporting faithfulness, answer relevancy, and context precision scores.

3. Suggested Technical Stack

You are not required to use this stack. Deviations are acceptable — explain them in your README.

COMPONENT	SUGGESTED OPTION
Data Ingestion	Crawl4AI / Playwright / BeautifulSoup4 + pdfplumber
Chunking	LangChain RecursiveCharacterTextSplitter (semantic) / LlamaIndex
Embeddings	OpenAI text-embedding-3-small OR Cohere embed-english-v3.0
Vector Store	Qdrant (self-hosted) / Pinecone / ChromaDB
Retrieval	Hybrid BM25 + Dense → MMR re-ranking
LLM	GPT-4o / Claude Sonnet 3.5 / Gemini 1.5 Pro
Framework	LangChain / LlamaIndex / custom
Guardrails Library	Guardrails AI / NeMo Guardrails / prompt-level enforcement
UI	Streamlit / Next.js / Gradio
Evaluation	RAGAS / custom test harness
Deployment	Docker (required) + Railway / Render / HuggingFace Spaces (optional)

API keys must never be hardcoded. Use .env files with a .env.example committed to the repository. Any submission containing exposed credentials will be disqualified immediately.

4. Evaluation Rubric

CRITERIA	WHAT WE ARE ASSESSING	POINTS
RAG Pipeline Quality	Chunk quality, retrieval relevance, embedding choice rationale, hybrid retrieval implementation	25
Knowledge Accuracy	Correct product and brand information from retrieved context; no hallucination; source citations present	20
Guardrails Implementation	All six MUST guardrails functional and demonstrable in the test suite	20
Code Quality & Architecture	Modular, readable code; proper environment variable handling; Docker; README completeness	15
UI & Demonstrability	Working interface, age gate, source display, streaming or loading state	10
Test Suite	15+ test queries with logged retrieved context, responses, and pass/fail outcomes	10
Bonus: RAGAS Evaluation	Automated faithfulness and context precision scores reported	+5
Bonus: Multi-turn Memory	Coherent session-level context retention across conversation turns	+5

5. Submission Requirements

The following deliverables are all required. Incomplete submissions will be scored accordingly.

GitHub Repository	Public or shared with hiring@markgrid.ai. Must include full commit history — do not submit a single ZIP dump.
README.md	Setup instructions, architecture decisions, stack rationale, known limitations, and instructions to run the test suite.
Dockerfile + docker-compose.yml	Must run end-to-end with a single 'docker compose up' command.
.env.example	All required environment variable names listed, with no real credentials committed.
Test Results Log	15+ queries with retrieved context, response, and pass/fail outcome — in Markdown or JSON format.
2-Minute Loom / Screen Recording	Live demo of the working chatbot including at least two guardrail triggers in action.

How to submit: Email hiring@markgrid.ai with subject line: RAG Assignment — [Your Full Name]. Include your GitHub repository link and Loom recording link in the email body.

6. Ground Rules

- You may use any LLM APIs (OpenAI, Anthropic, Google, Cohere, etc.). Include estimated API costs in your README if significant.
- Open-source models (Mistral, LLaMA, etc.) are permitted, but output quality is held to the same standard.
- AI coding assistants (Copilot, Cursor, Claude) are permitted. We assess architectural decisions and output quality, not keystrokes.
- This is an individual assignment. Sharing solutions with other candidates is grounds for immediate disqualification.
- Only public web pages may be scraped. Do not attempt to access Pernod Ricard's internal APIs, backend systems, or any non-public data.
- Questions may be directed to hiring@markgrid.ai. Responses will be provided during business hours.